DAY - 22 SEAT NUMBER

2026 III 10 1100 **V - 767** (E)

ELECTRONICS

PAPER - II (C-2)

Time : 3 Hours **4 Pages** **Max. Marks : 50**

Instructions : (1) *All questions are compulsory.*
(2) *Draw neat labelled diagrams wherever necessary.*
(3) *Figures to the right indicate full marks.*
(4) *Use of log table is allowed.*

1. (A) Select correct alternatives from the following sub-questions and rewrite the complete sentences :

(a) In any logic family when speed of operation is increased the total amount of power dissipated in it ________. 1

(i) Increases
(ii) Decreases
(iii) Does not change
(iv) Zero

(b) 1:64 Demultiplexer can be designed with ________ select lines. 1

(i) 4
(ii) 5
(iii) 6
(iv) 3

(c) In ________ A/D converter for high resolution system many comparators are necessary. 1

(i) Counter type
(ii) Successive approximation type
(iii) R-2R ladder type
(iv) Simultaneous type

(d) To produce graphical images ________ device is used. 1

(i) OMR

(ii) COM

(iii) Plotter

(iv) OCR

(B) Attempt **any two** of the following :

(a) With suitable example explain Hexadecimal to Decimal Conversion. 3

(b) Define : (i) Power dissipation, (ii) Fan-in, Fan-out, (iii) Noise margin. 3

(c) Explain the working of Master-slave JK Flip-flop. Draw suitable diagram. 3

2. (A) Attempt **any two** of the following :

(a) Solve : 3

(i) $(1110)_2 - (0101)_2 = (?)_2$

(ii) $(1101)_2 + (1011)_2 = (?)_2$

(iii) $(11010)_2 - (1010)_2 = (?)_2$

(b) Draw and explain basic circuit of CMOS Inverter. 3

(c) Draw the diagram of 4-bit right shift register using D-flip-flop and explain its working. 3

(B) Attempt **any one** of the following :

(a) State and prove Demorgans theorems. Draw logic diagram and truth table. 4

(b) Draw logic diagram of 4:1 Multiplexer using logic gates and explain it with truth table. 4

3. (A) Attempt **any two** of the following :

(a) Draw and explain basic circuit of TTL NOR gate. 3

(b) Implement the expression

$f(A,B,C) = \Sigma m\ (2, 5, 4, 7)$ Using multiplexer. 3

(c) Explain the working of counter type Analog to Digital Converter. 3

(B) Attempt **any one** of the following :

(a) Draw the logic diagram of 3-bit Synchronous Counter and explain its working. 4

(b) Draw block diagram of Digital Computer and explain function of each block. 4

SEAT NUMBER | | | | | | | |

4 (A) Attempt **any two** of the following

(a) Explain 1's complement and 2's complement with suitable examples. **3**

(b) Explain NAND and NOR gates in brief **3**

(c) Explain the working clocked RS Flip-flop using NAND gates **3**

(B) Attempt **any one** of the following

(a) Explain 1:4 Demultiplexer. Draw logic diagram, truth table. Give one application of Demultiplexer **4**

(b) Explain the working of R-2R ladder type Digital to Analog Converter. **4**

5 (A) Attempt **any two** of the following

(a) Explain Hexadecimal number system. Convert Hexadecimal number into binary **3**

(i) $(AB9)_{16} = (?)_2$

(b) Implement the following expressions using Demultiplexer having active low outputs **3**

(i) $f_1 = \Sigma m\ (1, 3, 5, 7)$

(ii) $f_2 = \Sigma m\ (2, 4, 6)$

(c) Explain : (i) RAM, (ii) EPROM, (iii) EEPROM **3**

(B) Attempt **any one** of the following

(a) Prove using Boolean Laws : **4**

(i) $\overline{A \oplus B} = A \cdot B + \bar{A} \cdot \bar{B}$

(ii) $A + \bar{A}B = A + B$

(b) What is Modulus of counter ? How many Flip-flops are required to construct each of following counter : **4**

(i) Mod - 8

(ii) Mod - 16

(iii) Mod - 32

OR

5 (A) Attempt any two of the following :

(a) What are different number systems ? Give Radix for various number system. 3

(b) Explain Half adder with logic diagram and truth table. 3

(c) Explain the working of clocked D Flip-flop with logic diagram and truth table. 3

(B) Attempt any one of the following :

(a) What is Decoder ? Explain the working of BCD to Decimal decoder using gates 4

(b) Explain the working of Weighted Resister type Digital to Analog Converter. Write its disadvantages. 4